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Risk Management

COMP1003 - Lecture 18

Dr Horia A. Maior



Today

- Meme of the Week
- Lecture
- Whats coming up



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Meme of the Week



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The Lecture

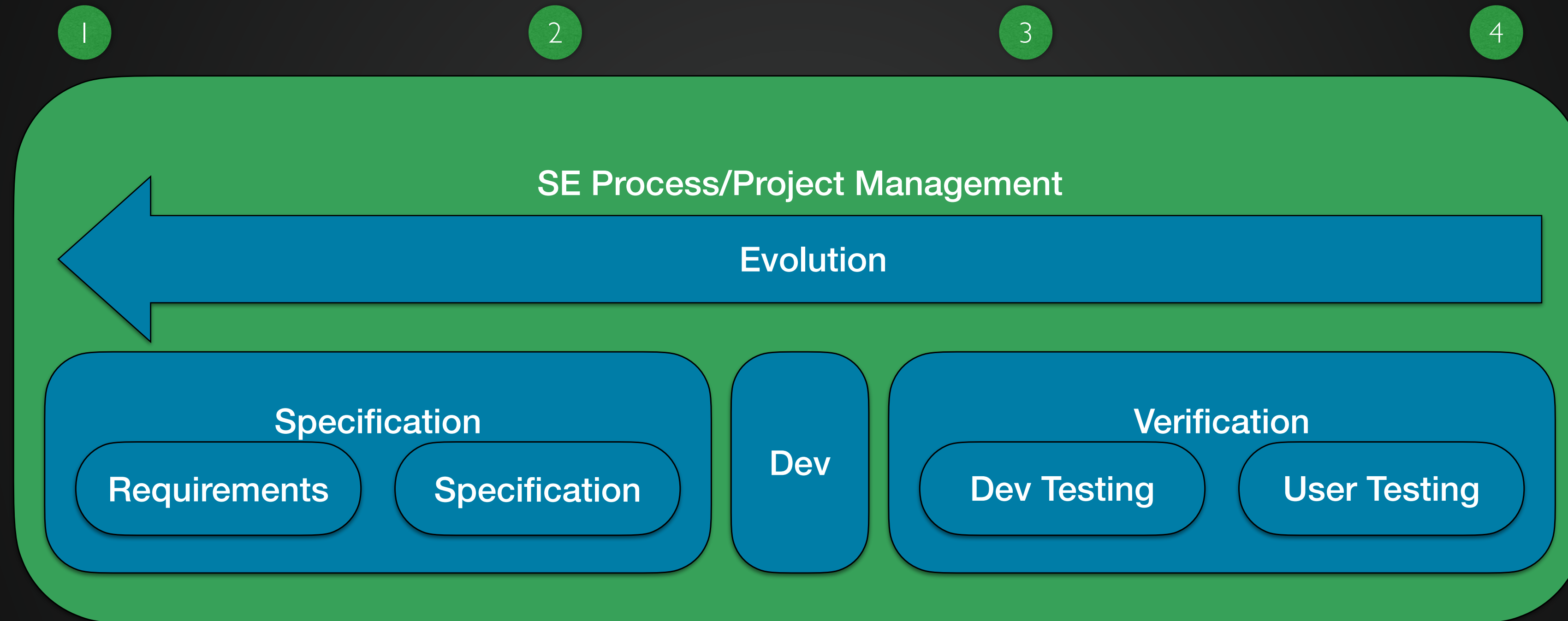


Process/Project Management

- Use good reliable methods 1
- Deliver good quality product 2
- **Reducing as much risk as possible** 3
- » plan / budget / manage a project based on these 4

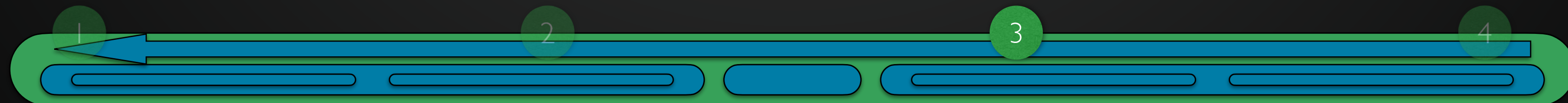


Keeping Track of This Module



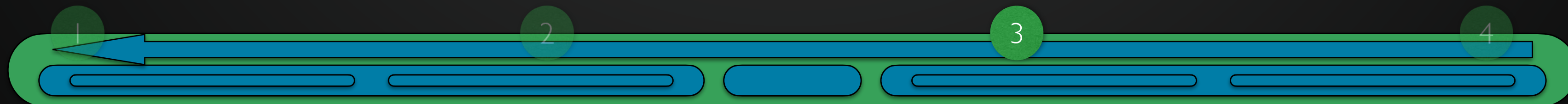


Keeping Track of This Module



Lecture 18 Learning Outcomes

- **Software Risks** » Security/Dependability » new reqs, specs, designs
 - so do a functional risk analysis early as possible
 - create some clear 'shall-nots'
- Most project planning is about managing **Project Risk**
 - mitigating, avoiding, or handling
 - so do a project risk analysis at the start
- Dont forget - Risk Monitoring - its not over after the planning





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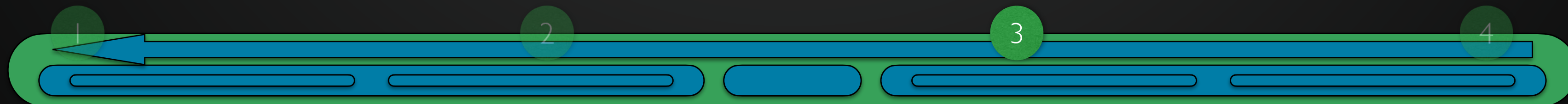
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Software Risks



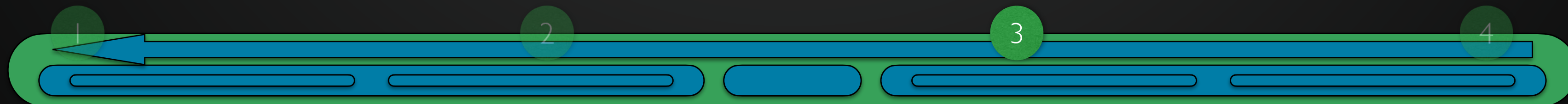
Risk Requirements for Software?

- Prison release software
- Hawaii Missile Launch Notification software
- Breast Cancer Scan invite software



Risk Requirements for Software?

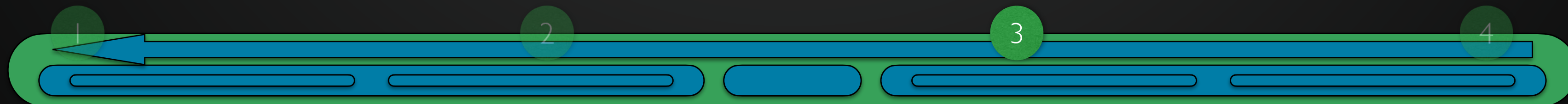
- Requirements & Specs were all about 'should' and 'shall'
- What about 'should not'?
 - we have to actively think about this
- What should definitely not happen with software?
 - people access other peoples finances
 - software doesn't break during critical processes
 - services go down for periods of time





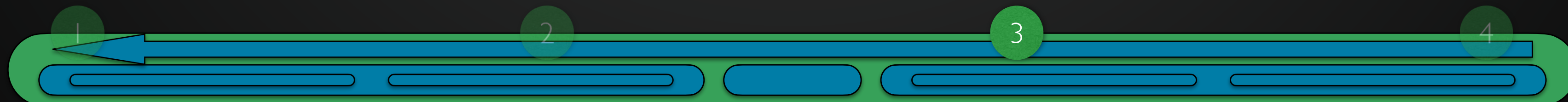
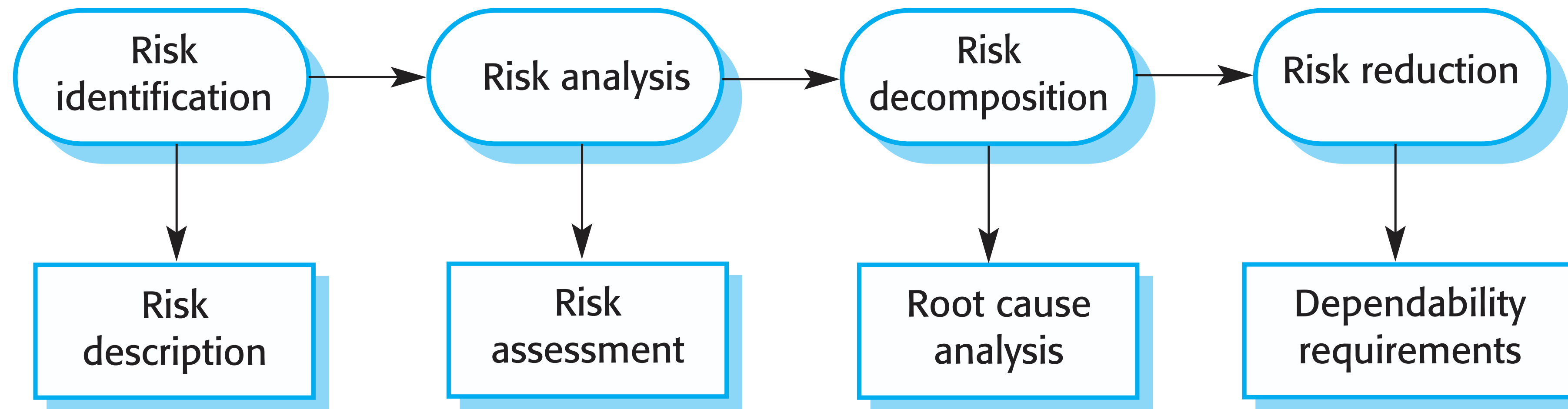
Shall-not Requirements

- New Functional Requirements
 - what should not happen
 - what should happen for non-correct usage or errors
- New Non-functional Requirements
 - define the reliability and availability of software



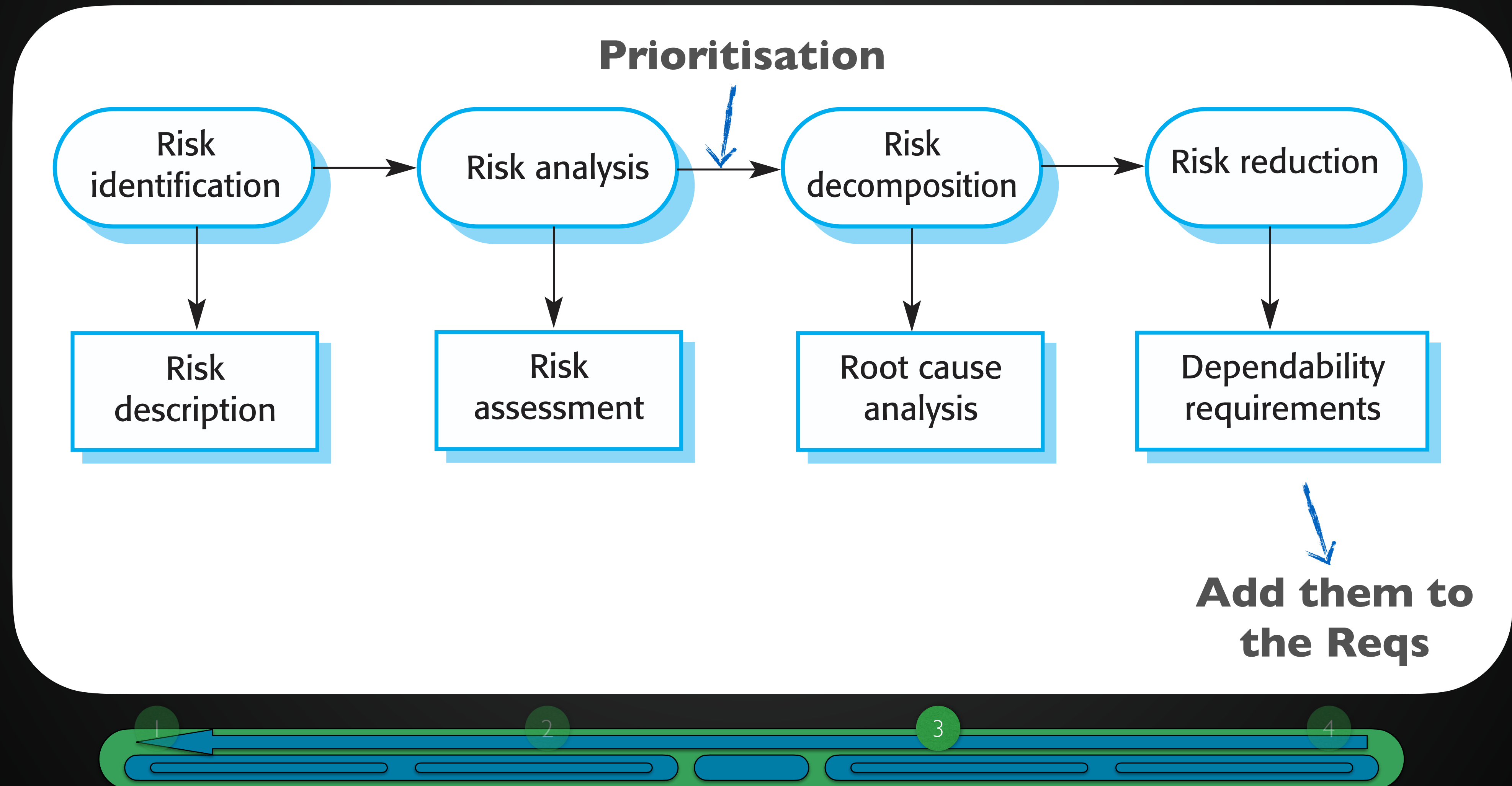


Risks » Requirements





Risks » Requirements





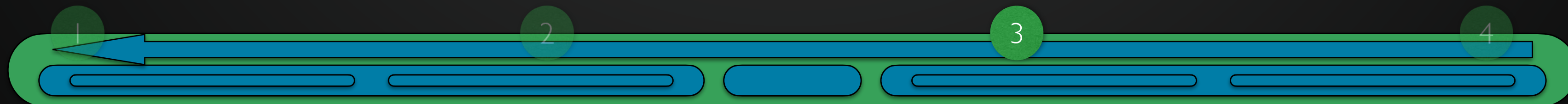
[Software] Risk Analysis

Identified hazard	H a z a r d probability	Accident severity	Estimated risk	Acceptability
1. Insulin overdose computation	Medium	High	High	Intolerable
2. Insulin underdose computation	Medium	Low	Low	Acceptable
3. Failure of hardware monitoring system	Medium	Medium	Low	ALARP
4. Power failure	High	Low	Low	Acceptable
5. Machine incorrectly fitted	High	High	High	Intolerable
6. Machine breaks in patient	Low	High	Medium	ALARP
7. Machine causes infection	Medium	Medium	Medium	ALARP
8. Electrical interference	Low	High	Medium	ALARP
9. Allergic reaction	Low	Low	Low	Acceptable

As Low As Reasonably Possible

[Software] Risk Decomposition

- There might be several ways that a risk can occur
- Sometimes best solution to solve a problem
 - is not the 'best for that problem'
 - but the best for all related problems
- If you collect shall-not specifications
 - you make it part of your collective design decisions
 - even better if not 'adjustments to the ideal design'
 - but designed to be appropriately risk adverse



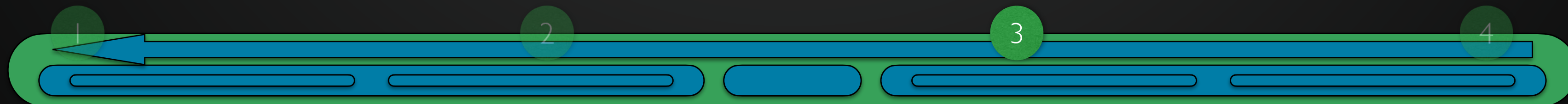


Fixes vs Good Solutions



[Software] Risk Reduction

- 1) Hazard Avoidance - so it cannot occur
- 2) Hazard Detection & Removal - so systems recover nicely
- 3) Damage Limitation - so the impact is limited
- These are design issues
 - and might affect the design in the first place



Risk Specification » Design

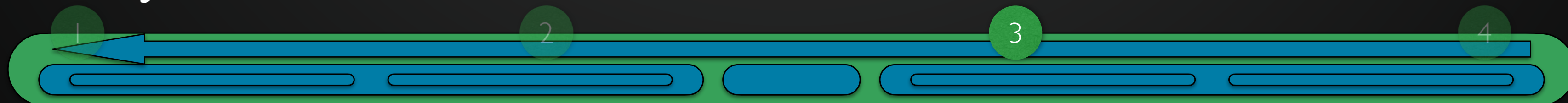
- SR1: The system shall not deliver a single dose of insulin that is greater than a specified maximum dose for a system user.
- SR2: The system shall not deliver a daily cumulative dose of insulin that is greater than a specified maximum daily dose for a system user.
- SR3: The system shall include a hardware diagnostic facility that shall be executed at least four times per hour.
- SR4: The system shall include an exception handler for all of the exceptions that are identified in Table 3.
- SR5: The audible alarm shall be sounded when any hardware or software anomaly is discovered and a diagnostic message, as defined in Table 4, shall be displayed.
- SR6: In the event of an alarm, insulin delivery shall be suspended until the user has reset the system and cleared the alarm.



Non-functional Risks

Failure type	Description
Loss of service	The system is unavailable and cannot deliver its services to users. You may separate this into loss of critical services and loss of non-critical services, where the consequences of a failure in non-critical services are less than the consequences of critical service failure.
Incorrect service delivery	The system does not deliver a service correctly to users. Again, this may be specified in terms of minor and major errors or errors in the delivery of critical and non-critical services.
System/data corruption	The failure of the system causes damage to the system itself or its data. This will usually but not necessarily be in conjunction with other types of failures.

- Unlike risks and hazards, reliability can be measured!
 - downtime - or uptime (availability)
 - rate of failures
 - recovery time



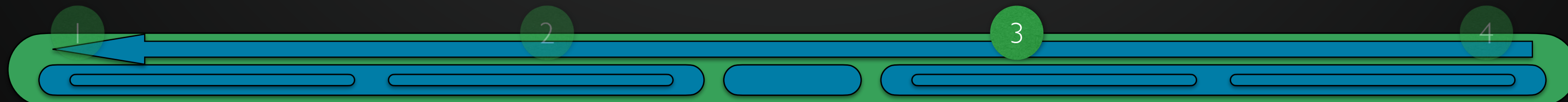


Availability Measures

Availability	Explanation
0.9	The system is available for 90% of the time. This means that, in a 24-hour period (1,440 minutes), the system will be unavailable for 144 minutes.
0.99	In a 24-hour period, the system is unavailable for 14.4 minutes.
0.999	The system is unavailable for 84 seconds in a 24-hour period.
0.9999	The system is unavailable for 8.4 seconds in a 24-hour period. Roughly, one minute per week.</TB>

- Interesting example about Google's uptime:

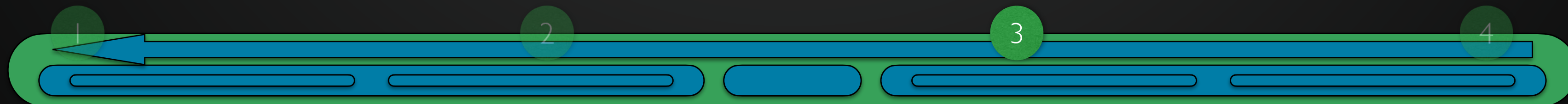
<http://www.wired.com/2016/04/google-ensures-services-almost-never-go/>





Risk-based Release Tests

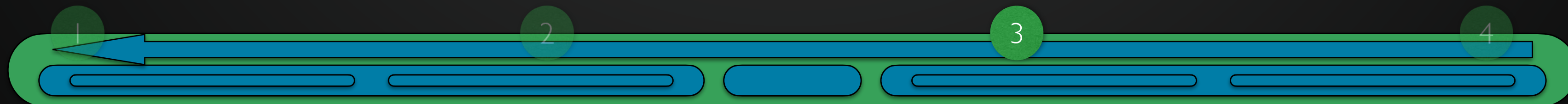
- Your test plan (for release or acceptance tests) might be driven by these
- It might dictate choice of hardware, or database software
- It might mean designing to optimise/minimise use of those





Security Concerns

- Approached differently
- What are the most valuable components in the project
 - credit card data?
 - customer patient data?
 - proprietary algorithms?
- Identify possible routes to this information
- Especially relevant for APIs, which purposefully give access





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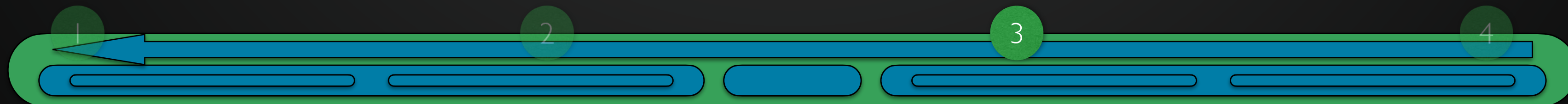
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Project Risks



Project Management Concerns

- 1) Deliver the software to the customer on schedule
- 2) Keep overall costs within budget
- 3) Deliver software that meets the customer's expectations
- 4) Maintain a happy and well-functioning dev team

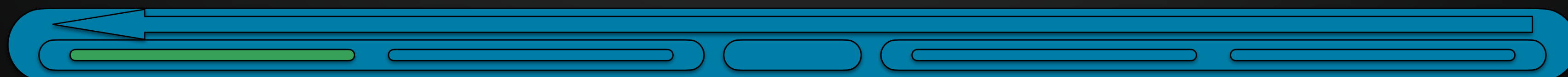




Ongoing SE Project Failures

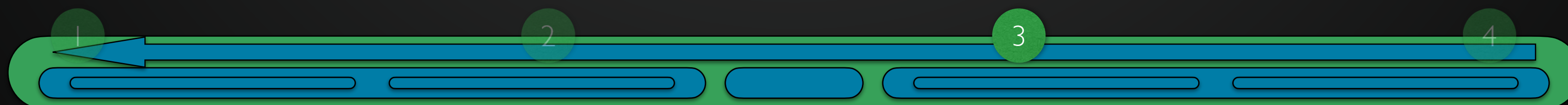
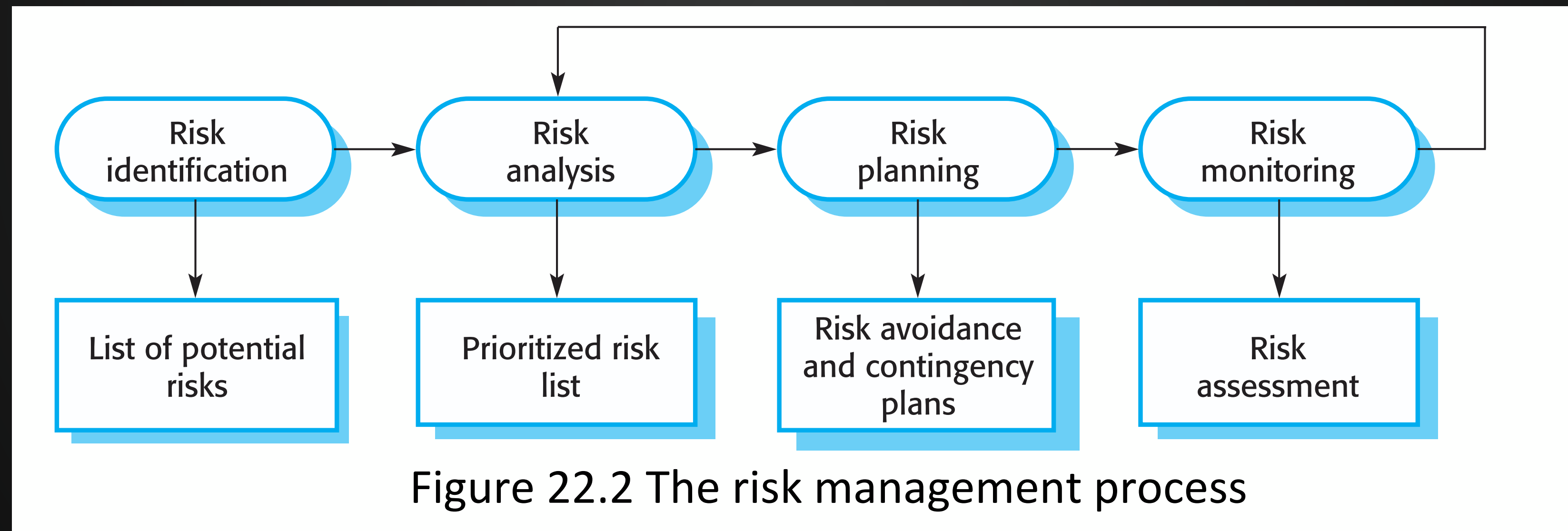
- The global cost of IT failure was estimated at + 6 trillion dollars in 2009
Source: Roger Sessions, The IT Complexity Crisis: Danger and Opportunity
- Only 32% of software projects were “successfully completed” (i.e., on time, on cost, and with expected functionality) in 2009
Source: Standish CHAOS 2009 Update
- Only 16% of software projects were successfully completed in the UK in 2009
Source: British Computer Society
- “A failing industry ...
 - *If building engineers built buildings with the same care as software engineers build systems, the first woodpecker to come along would be the end of civilization as we know it.”*

Source: Paul Dorsey, Top 10 Reasons Why Systems Projects Fail



[Project] Risk Analysis

- Risk Management is one of the most important jobs for a project manager
 - considering and preparing for possible problems in the future
 - for the project going smoothly (not software going wrong)
- All risks should be listed, and a strategy considered

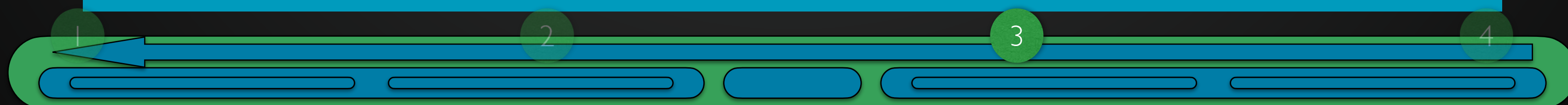




[Project] Risk Examples

Risk	Affects	Description
Staff turnover	Project	Experienced staff will leave the project before it is finished.
Management change	Project	There will be a change of organizational management with different priorities.
Hardware unavailability	Project	Hardware that is essential for the project will not be delivered on schedule.
Requirements change	Project and product	There will be a larger number of changes to the requirements than anticipated.
Specification delays	Project and product	Specifications of essential interfaces are not available on schedule.
Size underestimate	Project and product	The size of the system has been underestimated.
CASE tool underperformance	Product	CASE tools, which support the project, do not perform as anticipated.
Technology change	Business	The underlying technology on which the system is built is superseded by new technology.
Product competition	Business	A competitive product is marketed before the system is completed.

Figure 22.1 Examples of common project, product, and business risks

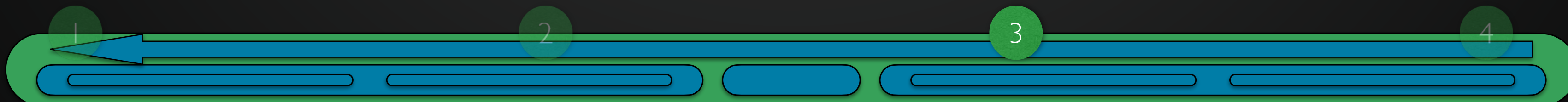




[Project] Risk Types

Risk type	Possible risks
Technology	The database used in the system cannot process as many transactions per second as expected. (1) Reusable software components contain defects that mean they cannot be reused as planned. (2)
People	It is impossible to recruit staff with the skills required. (3) Key staff are ill and unavailable at critical times. (4) Required training for staff is not available. (5)
Organizational	The organization is restructured so that different management are responsible for the project. (6) Organizational financial problems force reductions in the project budget. (7)
Tools	The code generated by software code generation tools is inefficient. (8) Software tools cannot work together in an integrated way. (9)
Requirements	Changes to requirements that require major design rework are proposed. (10) Customers fail to understand the impact of requirements changes. (11)
Estimation	The time required to develop the software is underestimated. (12) The rate of defect repair is underestimated. (13) The size of the software is underestimated. (14)

Figure 22.3 Examples of different types of risk

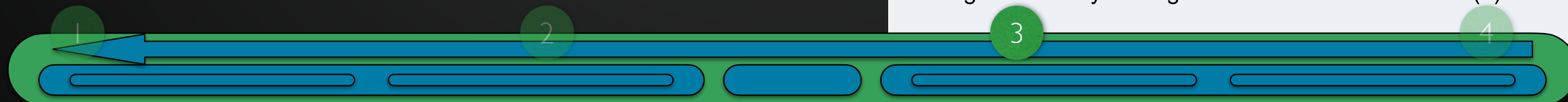




[Project] Risk Prioritisation

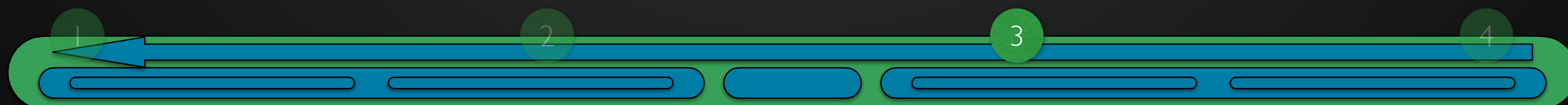
- Probability of it
 - very low (<10%)
 - low (10-25%)
 - moderate (25-50%)
 - high (50-75%)
 - very high (>75%)
- The effect of it!
 - catastrophic (project fail)
 - serious (expense/delays)
 - tolerable (in contingency)
 - insignificant (don't care)
- Both are important

Risk	Probability	Effects
Organizational financial problems force reductions in the project budget (7).	Low	Catastrophic
It is impossible to recruit staff with the skills required for the project (3).	High	Catastrophic
Key staff are ill at critical times in the project (4).	Moderate	Serious
Faults in reusable software components have to be repaired before these components are reused. (2).	Moderate	Serious
Changes to requirements that require major design rework are proposed (10).	Moderate	Serious
The organization is restructured so that different management are responsible for the project (6).	High	Serious
The database used in the system cannot process as many transactions per second as expected (1).	Moderate	Serious
The time required to develop the software is underestimated (12).	High	Serious
Software tools cannot be integrated (9).	High	Tolerable
Customers fail to understand the impact of requirements changes (11).	Moderate	Tolerable
Required training for staff is not available (5).	Moderate	Tolerable
The rate of defect repair is underestimated (13).	Moderate	Tolerable
The size of the software is underestimated (14).	High	Tolerable
Code generated by code generation tools is inefficient (8).	Moderate	Insignificant



[Project] Risk Planning

- You should choose a strategy for the risks
- **Avoidance strategies** - actions taken to reduce the risk happening
- **Minimization strategies** - reducing the impact if it happens
- **Contingency plans** - what you will do/change if it happens
- Avoidance is always the best strategy, if possible.
 - its often worth taking pre-emptive actions to avoid risks





[Project] Risk Strategies

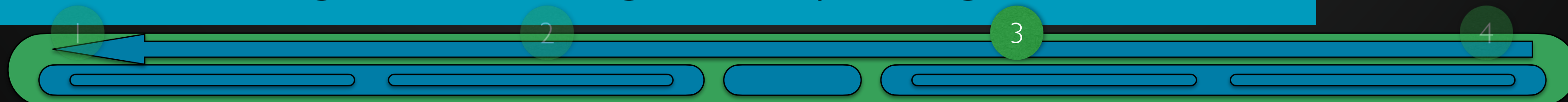
Risk	Strategy
Organizational financial problems	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business and presenting reasons why cuts to the project budget would not be cost-effective.
Recruitment problems	Alert customer to potential difficulties and the possibility of delays; investigate buying-in components.
Staff illness	Reorganize team so that there is more overlap of work and people therefore understand each other's jobs.
Defective components	Replace potentially defective components with bought-in components of known reliability.
Requirements changes	Derive traceability information to assess requirements change impact; maximize information hiding in the design.
Organizational restructuring	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business.
Database performance	Investigate the possibility of buying a higher-performance database.
Underestimated development time	Investigate buying-in components; investigate use of a program generator.

Minimisation

Avoidance

Contingency

Figure 22.5 Strategies to help manage risk

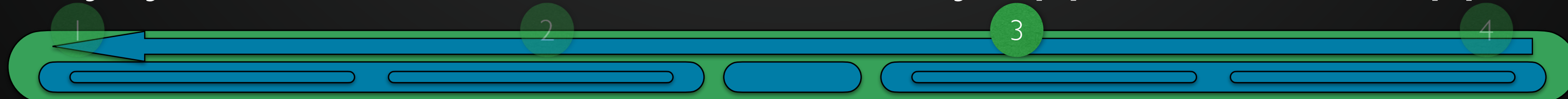




[Project] Risk Monitoring (!)

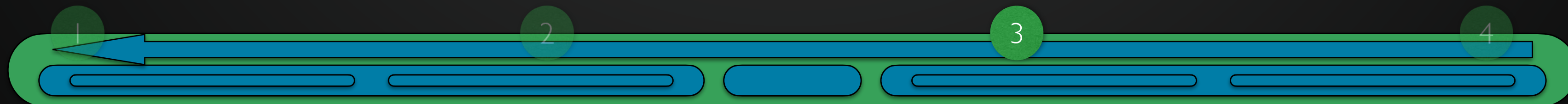
Risk type	Potential indicators
Technology	Late delivery of hardware or support software; many reported technology problems.
People	Poor staff morale; poor relationships amongst team members; high staff turnover.
Organizational	Organizational gossip; lack of action by senior management.
Tools	Reluctance by team members to use tools; complaints about CASE tools; demands for higher-powered workstations.
Requirements	Many requirements change requests; customer complaints.
Estimation	Failure to meet agreed schedule; failure to clear reported defects.

- Obviously, you want to check to see if any appear to be happening



Lecture 18 Learning Outcomes

- Software Risks » Security/Dependability » new reqs, specs, designs
 - so do a functional risk analysis early as possible
 - create some clear 'shall-nots'
- Most project planning is about managing Project Risk
 - mitigating, avoiding, or handling
 - so do a project risk analysis at the start
- **Dont forget - Risk Monitoring** - its not over after the planning
- Up Next! The actual planning.





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Coming Up



Week Starting	Lecture & Lab Slots		
	Monday 9-10am Business B52	Thursday 4-6pm A32/B52/C11/C50	Friday 4-5pm Business B52
1) 27 Jan	Module Intro	No Lab	Git Ready for Teams
2) 3 Feb	Requirements Engineering	Team Activity: Git and Markdown	Requirements Gathering
3) 10 Feb	Requirements Models	Team Activity: UML Intro	Requirements Validation
4) 17 Feb	Specification	Team Activity - More UML	Prototyping
5) 24 Feb	Code Planning	Team Activity - Requirements Gathering	Coding in Teams
6) 3 Mar	Test-Driven Development & JUnit Demo	Team Activity - JUnit	Release Testing
7) 10 Mar	Config & Deployment	Team Activity - Advanced Git	Advanced Git Demo
8) 17 Mar	Software Maintenance	CW start assessed 50% of the module	Industry Guest Lecture?
9) 24 Mar	Overview 1: Traditional vs Agile	CW support continued	Overview 2: Software Quality
E A S T E R			
10) 28 Apr	Overview 3: Risk Management	CW Support continued	Project Planning & Management
11) 5 May	Bank Holiday (No Lecture)	CW Delivery Friday the 9th of May at 12 noon (ideally end of lab on Thursday)	No Lecture
12) 12 May	No Lecture Individual Activity (at home) - PERT/Gantt	No Lab	No Lecture



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CW



- You do not have to stick to your CW plan
 - You were just supposed to realise that planning is hard, but hopefully that testing is better from having done it.
- Check the CW assessment criteria (in the briefing document)
 - Quality and coverage of tests
 - How well you code as a team, inc good use of version control, and how you improve bad code
 - Quality of the report (which is an evolved project plan)
- Most of the marks are how comprehensively you test the 3 classes
- Need help? Remember you have to use git issues



Handling absent team members

- Make it clear in the code and markdown who is doing what
- Do your bit well
- Try to do a vertical slice (a bit of everything, but not all of everything)
- Peer assess accurately
- Don't do their work, or they'll get a good mark and you'll get tired



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Q&A